

RAHAT IZHAR, Ph.D.

Software Engineering | Artificial Intelligence | Intelligent Systems | Academic Leadership

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ACADEMIC PROFILE

Computer Engineering Ph.D. and university Lecturer/Head of ICT with an academic profile spanning Software Engineering, Artificial Intelligence, Machine Learning, Natural Language Processing, Data Science, computing education, research supervision and programme leadership. Requirements Engineering is a core research foundation, while current and emerging work extends into software analytics, trustworthy AI, computer vision, applied intelligent systems, healthcare, agriculture and decision support. Experience combines peer-reviewed research, broad undergraduate and MBA-level teaching, curriculum and assessment design, quality assurance, student/postdoctoral mentoring, peer review and international academic engagement.

CURRENT ACADEMIC APPOINTMENTS & LEADERSHIP

Head of ICT Department

25 Jun 2026 - Present

British International College, Krirk University, Bangkok, Thailand

- Lead ICT academic coordination, curriculum implementation, faculty support, assessment alignment, examination moderation, programme documentation and continuous-improvement activities.
- Support student progression, technical/final-year projects, internships, recruitment and programme-level academic engagement.

Lecturer

01 Aug 2024 - Present

British International College, Krirk University, Bangkok, Thailand

- Teach undergraduate and MBA-level subjects across programming, OOP, algorithms, networks, web development, AI-IoT, systems analysis, data science and Information Systems for Business Management.
- Design course outlines, learning materials, practical activities, assignments, examinations, rubrics and LMS-supported learning for international cohorts.

EDUCATION

Ph.D. in Computer Engineering | *Chiang Mai University, Thailand* | 21 Jun 2021 - 13 Jun 2025

Doctoral research on AI/NLP-enabled Requirements Engineering in Agile software development. Thesis: "Analytical Impacts of Requirements Management in Agile Methodology." Recipient of the Chiang Mai University Presidential Scholarship (2021-2024).

M.S. in Software Engineering | *Bahria University, Islamabad, Pakistan* | 06 Sep 2017 - 06 Sep 2019

Graduate focus included Requirements Engineering, Software Testing & Quality Assurance, databases, software metrics, project management and object-oriented software design. Research addressed ERP requirements elicitation and customization.

B.S. in Computer Science | *SZABIST, Pakistan* | 03 Sep 2012 - 30 Aug 2016

Computer Science foundation in programming, software development and information technologies. Final-year project: intelligent job recommendation system with Java MVC backend and Android frontend.

TEACHING, CURRICULUM & ACADEMIC LEADERSHIP

- Teaching portfolio: Basic Programming; Object-Oriented Programming; Data Structures & Algorithms; Computer Networks; Web Design & Development; AI & IoT; Information Systems Analysis & Design; Fundamentals of Data Science & Data Analytics; MBA Information Systems for Business Management.

- Teaching approach: structured conceptual explanation combined with live coding, worked examples, data analysis, simulation, project-based learning, case studies and guided analytical practice.
- Curriculum and quality: course design, learning outcomes, assessment design, rubrics, examination moderation, programme documentation, quality-assurance evidence and continuous improvement.
- Supervision and mentoring: undergraduate technical/final-year projects and current postdoctoral research mentoring with emphasis on reproducibility, evaluation quality and publication development.

RESEARCH PROGRAMME & SCHOLARSHIP

- Software Engineering and requirements intelligence: requirements classification, ambiguity analysis, semantic grouping, traceability, prioritization, meaning-preserving refinement and software-quality analytics.
- AI, NLP and LLM-enabled systems: machine learning, semantic similarity, transformer-based representations, language intelligence and AI-assisted software-engineering workflows.
- Trustworthy AI: explainability, robustness, generalization, ablation, metamorphic testing, failure analysis and reproducible empirical evaluation.
- Applied intelligent systems: interdisciplinary work in biomedical imaging, agriculture/plant-disease robustness, logistics/cold-chain decision support and other data-driven application areas.

SELECTED RESEARCH PROJECTS & CURRENT DIRECTIONS

HLSRP - Meaning-Preserving Requirements Refinement | *Software Engineering / NLP*

Hybrid linguistic-semantic refinement of natural-language requirements using sentence embeddings, semantic-drift controls and external evaluation.

Beyond Random Splits | *Software Analytics*

Multi-axis generalization audit of bug-report resolution prediction across lexical and semantic representations, emphasizing realistic evaluation beyond random train-test splitting.

Explainable Cold-Chain Decision Support | *Applied AI*

Machine-learning failure prediction with explainability and LLM-assisted recovery recommendations for operational logistics settings.

Metamorphic Testing for Plant-Disease Classification | *Agricultural AI / Robustness*

Current supervised work on robustness evaluation under lighting, blur, rotation, occlusion and background changes using baseline deep-learning models and metamorphic relations.

PEER-REVIEWED PUBLICATIONS

1. R. Izhar, S. N. Bhatti, and S. A. Alharthi, "Bridging Precision and Complexity: A Novel Machine Learning Approach for Ambiguity Detection in Software Requirements," *IEEE Access*, vol. 13, pp. 12014-12031, 2025. DOI: 10.1109/ACCESS.2025.3529943.
2. R. Izhar, K. Cosh, L. Ramingwong, S. Ramingwong, and S. N. Bhatti, "An Efficient Methodology for the Categorization of Software Requirements Using Natural Language Processing and Similarity Analysis," *IEEE Access*, vol. 13, pp. 83591-83606, 2025. DOI: 10.1109/ACCESS.2025.3568504.
3. R. Izhar, S. N. Bhatti, and S. A. Alharthi, "An NLP-Enabled Approach to Semantic Grouping for Improved Requirements Modularity and Traceability," *International Journal of Advanced Computer Science and Applications*, vol. 16, no. 2, 2025. DOI: 10.14569/IJACSA.2025.0160252.

4. R. Izhar, K. Cosh, and S. N. Bhatti, "Enhancing Agile Software Development: A Novel Approach to Automated Requirements Prioritization," 21st International Joint Conference on Computer Science and Software Engineering (JCSSE), IEEE, 2024, pp. 286-293. DOI: 10.1109/JCSSE61278.2024.10613648.
5. R. Izhar, S. N. Bhatti, S. Izhar, and A. M. Jadi, "Impact and Challenges of Requirements Management in Enterprise Resource Planning (ERP) via ERP Thesaurus," International Journal of Advanced Computer Science and Applications, vol. 9, no. 8, 2018. DOI: 10.14569/IJACSA.2018.090832.
6. M. Hayat, R. Izhar, M. Nadeem, H. Anjum, A. Muhammad, and S. Bhattacharjee, "HAMSRNet-Hybrid Attention Multiscale Super-Resolution Network for Endoscopic Images," ICoDT2, IEEE, 2025. DOI: 10.1109/ICoDT269104.2025.11360731.
7. M. Hayat, A. Dhaliwal, M. M. U. Din, R. Izhar, M. Nadeem, and N. Ahmad, "Cross-Attention Patch Fusion for Few-Shot Colorectal Tissue Generation," ICoDT2, IEEE, 2025. DOI: 10.1109/ICoDT269104.2025.11360745.
8. R. K., R. Izhar, and S. N. Bhatti, "Examining the Role of AI Experience in Shaping Consumer Attitudes and Adoption Behavior in Social Media Marketing," AIIT, IEEE, 2025. DOI: 10.1109/AIIT63112.2025.11082958.
9. W. Hong, C. Cao, S. N. Bhatti, and R. Izhar, "Artificial Intelligence Human Resource Management Ethic Concerns among Middle-level Managers in High-Tech Corporations," AIIT, IEEE, 2025. DOI: 10.1109/AIIT63112.2025.11082883.
A. Muhammad, M. Nadeem, H. Anjum, R. Izhar, S. Bhattacharjee, N. Ahmad, and M. Hayat, "Design of a Dual-Element MIMO Antenna Incorporating Neutralization Line for Isolation Enhancement in the 20-28 GHz Range," ICIT, IEEE, 2026. DOI: 10.1109/ICIT68548.2026.11577707.
10. R. Izhar, K. Cosh, and S. N. Bhatti, "Hybrid Modelling Prototype for the Requirements Prioritization in Agile," IWBDIAA 2023, Kenitra, Morocco, 2023.

SELECTED ACCEPTED / CURRENT RESEARCH PIPELINE

- "An Explainable AI and LLM-Based Work Logistics Framework for Cold-Chain Shipment Failure Prediction and Recovery Decision Support" - accepted for ICL 2027.
- "Beyond Random Splits: A Multi-Axis Generalization Audit of Bug Report Resolution Prediction Across Lexical and Semantic Representations" - journal manuscript under review, 2026.
- "HLSRP: A Hybrid Linguistic-Semantic Framework for Meaning-Preserving Refinement of Natural Language Requirements" - submitted 2026.

ACADEMIC SERVICE & INTERNATIONAL ENGAGEMENT

- Peer reviewer: ACM Transactions on Software Engineering and Methodology (TOSEM), Information Processing & Management, e-Informatica Software Engineering Journal, and IEEE Horizons of Information Technology and Engineering (HITE) 2026.
- Invited Resource Speaker, Universitas PGRI Madiun, Indonesia - "Cybersecurity Awareness," 11 June 2026.
- Invited Speaker, ICONIES 2025, Indonesia - "Harnessing Artificial Intelligence to Advance ESG and Islamic Economic Values."
- Invited Expert Speaker, GL Bajaj Institute of Management & Research, India - "Data Modelling Techniques in a Global Context," February 2025.
- Presenter, IEEE JCSSE 2024, Phuket, Thailand; international student skills-competition mentoring and academic outreach activities.

EARLIER ACADEMIC & INDUSTRY EXPERIENCE

English / Computing / Mathematics Teacher & Teacher Coordinator | *Laohajit Wittaya School, Thailand* | 14 Nov 2019 - 17 May 2021

Taught varied learner groups and supported classroom coordination in physical and hybrid learning environments.

Teaching & Research Assistant | *Bahria University, Pakistan* | Nov 2017 - Oct 2019

Supported Software Engineering, C++/Java Programming and Web Engineering laboratories and contributed to ERP requirements research.

Business Analyst | *4SGlobal Pvt. Ltd., Pakistan* | Oct 2017 - Jan 2018

Analysed business workflows, translated stakeholder needs into technical specifications and contributed to ERP/reporting work.

Junior Executive IT Officer | *SyntecX, Pakistan* | 16 Aug 2017 - 03 Oct 2017

Worked on full-stack web development using PHP, HTML/CSS, JavaScript, Bootstrap and APIs in an Agile environment.

Software Development Intern | *SyntecX, Pakistan* | 16 Jan 2017 - 16 Jul 2017

Contributed to web/application development and practical software-engineering workflows.

Data Warehouse / IT-BSS Intern | *CMPAK Zong HQ, Pakistan* | 06 Jun 2016 - 06 Sep 2016

Supported BI planning, data analysis and performance reporting using Oracle, Teradata SQL and IBM Cognos.

IT-System Development Intern | *Zarai Taraqiat Bank Ltd. (ZTBL), Pakistan* | 15 Jun 2015 - 29 Jul 2015

Developed a C#/.NET and Oracle-based contact directory, automated Excel reporting and supported database tasks.

TECHNICAL & PROFESSIONAL COMPETENCIES

AI / Data / Research | *Python, scikit-learn, NLP, Transformers, semantic similarity, Pandas, NumPy, SHAP, robustness evaluation*

Machine learning, language intelligence, explainability, generalization testing, empirical evaluation and reproducible research.

Software Engineering & Development | *Requirements Engineering, SDLC, testing, system analysis, Java, C/C++, C#, JavaScript, PHP, SQL, HTML/CSS, REST/API*

Software analysis and design, programming, web/application development and data-oriented systems.

Academic Practice | *Curriculum design, assessment, quality assurance, supervision, peer review, research writing, academic leadership*

Programme coordination, international teaching, student mentoring and scholarly service.

LANGUAGES

English - Full professional proficiency | Urdu - Native | Thai - Basic conversational understanding